

Harmful Manipulation and Election Security

The Capability-Deployment-Effect Gap

How frontier AI can turn political influence into infrastructure—and how to keep that power contestable.

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Status. Provisional findings and interpretations; subject to correction. Parts I–III retain the 28 August 2026 data freeze. Part IV uses the 1 September source-linked review. This revision adds editorial clarification and targeted source checks, not new experiments, independent replication, or upgraded Atlas evidence flags.

Claim boundary. The programme does not establish live deployment against an audience, human persuasion, agency transfer, or an AI-attributable election outcome. External studies support specified human or technical outcomes, not an evaluated package of democratic protection.

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Abstract

Frontier AI may threaten democracy without being shown to have changed an election. The concern is that models joined to personal data, trusted interfaces, distribution, and feedback can make influence persistent and adaptive, while concentrating control over both decisions and the evidence needed to scrutinise them.

This paper develops the Capability-Deployment-Effect Gap, or CDE Gap, to distinguish capability, configured systems, deployment, exposure, human and institutional effects, and electoral consequences. Its central concept is agency transfer: an affected person or institution loses practical, reflective, and contestable control as the system's controller gains power over the decision process.

Four research strands support a provisional governance synthesis. Exploratory records show campaign-planning assistance, not a deployed operation. The evaluation review identifies useful instruments but no shared, validated end-to-end measure of harmful manipulation across deployed systems. Election records document operations and harms without identifying a causal AI-attributable electoral outcome. A source-linked review of 118 policy implementations finds usable evidence for specific human and technical outcomes, alongside mismatches between what controls measure and the democratic protection attributed to them.

The implication is to govern evidenced mechanisms without claiming unmeasured downstream benefits. Apply existing duties within scope; use controls matched to tested outcomes; make consequential authority inspectable and revocable; and test whether people can challenge, switch, and continue without the system. Defensive monitors require the same scrutiny. These recommendations are not an evaluated protection package. The democratic objective is to preserve citizens' and institutions' capacity to understand, direct, refuse, and contest AI-mediated influence.

The argument in brief

"you shall know the truth and the truth shall make you free"

- John 8:32

The next Cambridge Analytica may look less like a consultancy than an ordinary service: an assistant that knows its users, mediates their information, and learns from their responses.

The central hypothesis is not irresistible persuasion. It is concentrated control: influence becomes infrastructure when the same actor can shape repeated interactions, observe their consequences, and decide what outsiders may inspect.

The CDE Gap separates what is technically possible, what is deployed, and what changes. Agency transfer asks who loses practical control and who gains it. Together they support action against an evidenced mechanism without claiming that a safeguard has protected an election.

The same test applies to defence. A safer agent may still serve a powerful operator. A warning may change a decision without making it better understood or easier to contest.

Table 1. Four research lenses on the influence pathway. These are separate bodies of evidence, not observations of one completed operation. Source: author synthesis of Guerrero 2026a–f and the cited research.

Research lens	Evidence reviewed	Governance lesson	Claim ceiling
I. Operations	Exploratory planning assistance and limits on experimental control. The \$10 budget question is unverified.	Govern exact routes, configurations, permissions, persistence, and evidence custody—not model names alone.	No deployed operation, human-effect estimate, or audited cost-to-effect claim.
II. Evaluation	Useful but non-equivalent instruments; incomplete and invalid pipeline observations.	Evaluate configured systems with human-calibrated outcomes. No shared validated end-to-end measure was identified.	No model ranking, longitudinal trend, human-effect estimate, or election effect.
III. Field	Documented attempts, distribution, harms, and institutional responses.	Preserve delivery denominators and separate occurrence, attribution, exposure, response, and electoral consequence.	No prevalence estimate, causal zero-effect claim, durable agency-transfer estimate, or AI-attributable national-election outcome.
IV. Policy	External studies on human and technical outcomes, delivery, and whose authority is protected.	Separate binding duties, tested component effects, and structural hypotheses. Evaluate coverage, timing, beneficiary, refusal, and functional exit.	No universal efficacy ranking or demonstrated durable agency preservation. External findings are not fellowship experiments.

The paper uses five evidence labels. Established evidence is directly supported by the cited record. Strong inference joins established facts through a well-supported mechanism but remains unobserved at the endpoint. Plausible hypothesis identifies a testable pathway with material missing links. Speculative scenario describes a consequential possibility whose joined conditions have not been shown. Open question marks what the present design or record cannot resolve.

Guerrero (2026a–f) identifies this programme’s protocols, records, data, and coding—not independent corroboration. The references separate research literature, public records, provider documentation, and programme materials. External research is not necessarily independent of AI providers; study design, affiliations, and source status still matter.

1. The next Cambridge Analytica may look ordinary

Cambridge Analytica is remembered as a story about psychographic persuasion. The regulatory record established deceptive data collection and failures of privacy and accountability, not a clean estimate of votes changed (Information Commissioner’s Office 2018; Federal Trade Commission 2019). The lesson for this paper is architectural: political purpose, personal data, targeting, and distribution can be assembled before those affected can inspect or challenge their use.

A future operation need not resemble a notorious consultancy. It could join an audience database, model, assistant, advertising account, messaging channel, and analytics dashboard. The democratic question is who controls their combination—and whether outsiders can reconstruct what happened before the decision window closes.

Frontier AI could make that combination easier to sustain: generation, personalisation, translation, and adaptation can be connected across repeated exchanges. Whether this reduces the expertise, coordination, or full cost of an effective operation remains an empirical question. This programme does not observe such a closed loop against real people.

The question is not only whether AI can persuade. It is who can join capability, data, identity, distribution, feedback, and evidence —and what power follows.

Two errors obscure that question. One leaps from an impressive output to an altered election. The other treats the absence of a vote-effect estimate as evidence that nothing important happened. Both overlook the deployed system between capability and outcome.

Impersonation, unlawful targeting, excessive action authority, and lost evidence can warrant intervention without a demonstrated electoral effect. The CDE Gap is designed to distinguish those grounds for action from claims about votes changed.

2. Influence, manipulation, and agency transfer

Influence is the wider category; manipulation is not a synonym for political persuasion. Separate the actor's intention, the technique, and the effect. A manipulative attempt can fail, a true claim can be delivered through impersonation, and dependency can develop without changing a political belief.

Persuasion preserves reflective agency when people can examine arguments, identify relevant interests, access alternatives, and refuse. Changing one's mind is not itself a loss of agency.

Manipulation undermines those conditions through deception, concealed purpose, exploitation of vulnerability or asymmetric knowledge, emotional coercion, dependency, or control over the decision environment. The test is impaired reflective and contestable choice, not simply successful influence (Susser, Roessler, and Nissenbaum 2019; El-Sayed et al. 2024).

The EU General-Purpose AI Code of Practice supplies a related, but distinct, regulatory category. Its Safety and Security chapter identifies harmful manipulation as a specified systemic risk, including strategic distortion of behaviour or beliefs through persuasion, deception, or personalised targeting. It highlights multi-turn and difficult-to-detect interactions in provider risk management under Article 55 (European Commission 2025b, 2026a).

The Code is neither a universal legal definition nor a standalone prohibition. The paper's normative test, the AI Act's Article 5 prohibitions, and the Code's systemic-risk category have different purposes and thresholds; Appendix E keeps them separate.

Agency transfer is the paired shift at the centre of this paper: a person, public, or democratic institution loses practical, reflective, and contestable control over attention, preference formation, choice, dependency, or institutional decisions, while the actor controlling the system gains power over that process.

This excludes ordinary, revocable delegation. Assistance can expand agency when users retain the ability to understand, direct, refuse, correct, challenge, and leave it. Neither agreement with an AI nor successful task completion alone establishes agency transfer.

Agency transfer can occur at four levels. At the micro level, a system may exploit private data, emotional state, trust, or cognitive limits. At the meso level, assistants, recommenders, and agents may become indispensable intermediaries. At the macro level, dominant interfaces may gain agenda-setting power, epistemic authority, or privileged access to institutions, extending wider concerns about automated authority and power (Lazar 2024). At the democratic level, these shifts can weaken autonomy, consent, pluralism, accountability, contestability, legitimacy, electoral integrity, and collective self-government.

This paper uses catastrophic risk narrowly: a persistent, society-scale degradation of democratic self-correction—the ability to form preferences under plural conditions, identify responsible power, contest political claims, replace governments, expose abuse, distinguish authentic from synthetic authority, and repair institutional error. The current evidence does not show that threshold has been crossed.

Agency transfer is the central analytical test, not a measured programme outcome. The empirical work examines enabling conditions and possible points of interruption.

3. The Capability-Deployment-Effect Gap

Capability, deployment, and effect organise the evidence; they are not a universal linear ladder. An impersonation can burden an institution without persuading a voter. A recommender can shift attention without changing a stated belief. Human and institutional effects can diverge and feed back into later deployment.

The fuller pathway includes the model, served configuration, controlling actor, purpose, data, permissions, distribution, authentic exposure, human or institutional response, and adaptive feedback.

Capability is not deployment; deployment is not effect

The four research parts observe different nodes and transitions in one risk system.

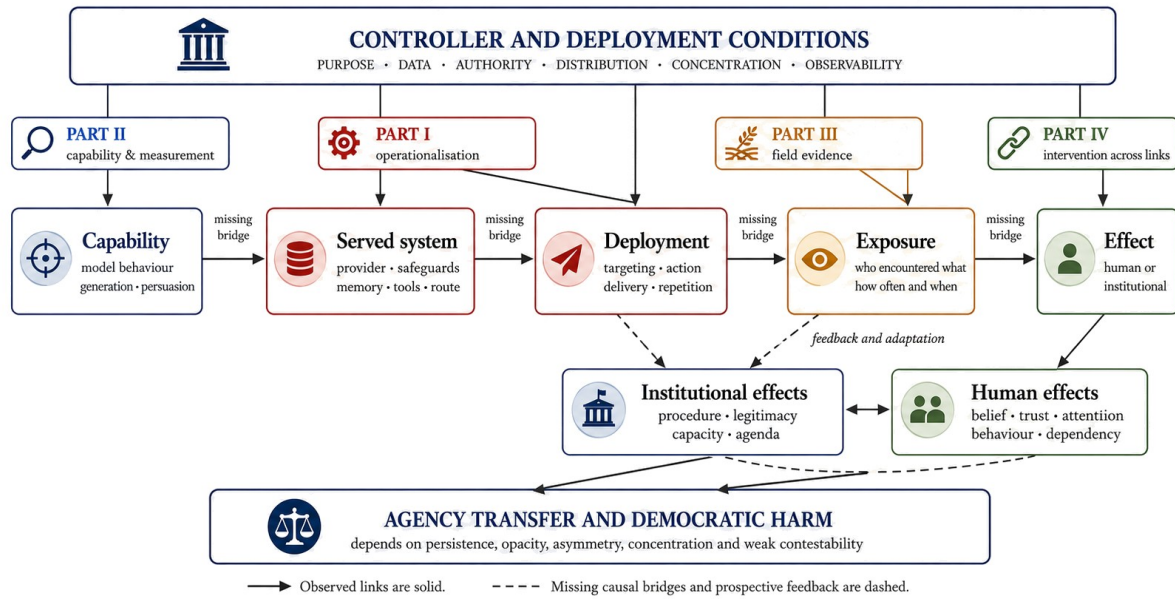


Figure 1. The Capability–Deployment–Effect Gap. Solid links indicate relations observed or directly studied in separate records or settings; they do not establish one completed causal chain. Dashed links indicate missing bridges or prospective feedback. Institutional harm need not pass through voter persuasion. Source: author’s conceptual synthesis.

The causal question is concrete: which AI capability is controlled by which actor, through which influence vector, against which target, producing what change in belief, preference, attention, trust, behaviour, or dependency—and with what transfer of agency, concentration of power, democratic harm, and possible mitigation?

Each transition needs its own evidence. A plan is not an executed action; a transmitted message may never be attended to. Counts of views are not counts of people, and changed beliefs among recipients are not an aggregate electoral effect.

The same discipline applies upstream. An intervention must be specified, reached, saved, independently reloaded, compared, and replicated. Appendix B’s G0–G7 record formalises those steps. Failure to save the intervention is a failure of experimental control, not a behavioural null.

A claim ceiling states the strongest conclusion supported by a result, its system configuration and comparator, and the evidence missing for the next claim. A policy-effect ceiling asks the same question of a safeguard: which endpoint improved, without promoting that endpoint into autonomy, resilience, or election integrity (Dassanayake et al. 2025).

These ceilings constrain claims, not all action. A telecom provider can address spoofed caller identification without first estimating votes changed. The action still needs an appropriate legal basis and proportionate scope; its public justification must stop where its evidence stops.

4. What the evidence says, and what it does not

The literature does not support complacency. It also does not support a straight line from persuasive text to democratic collapse.

Controlled studies show that language models can change reported attitudes in bounded settings. Personalised AI-generated messages have outperformed non-personalised alternatives in some experiments. GPT-4 has performed strongly against human opponents in short debates when given basic information about the person it addressed. Other studies report small but measurable shifts in policy attitudes and candidate preferences following AI-mediated dialogue (Matz et al. 2024; Salvi et al. 2025; Bai et al. 2025; Lin et al. 2025). A large harmful-manipulation study found that a tested model could produce manipulative behaviour and induce experimental changes in belief or behaviour, while manipulation propensity did not consistently predict efficacy (Akbulut et al. 2026).

These findings establish neither a universal effect nor universal superiority to humans. A 2025 meta-analysis found no significant overall persuasive advantage for LLMs and substantial variation across models, domains, and interaction designs (Hölbling, Maier, and Feuerriegel 2025). Personalisation, argument quality, repetition, and access to attention should be tested separately.

Conventional campaigns supply a restrictive prior. A synthesis of 49 field experiments found average persuasive effects near zero in general elections. Eady et al. found concentrated exposure to the Russian Internet Research Agency and no meaningful relationship with attitudinal or behavioural change in their linked sample. Other studies document rare and concentrated exposure to untrustworthy websites. These findings constrain generalisation; they do not prove zero effect (Kalla and Broockman 2018; Eady et al. 2023; Guess, Nyhan, and Reifler 2020; Moore, Dahlke, and Hancock 2023).

Chen et al. (2026) distinguish receiving a message from accepting it. Their experiments found chatbots about as persuasive as campaign advertising conditional on engagement, while securing conversations at scale remained a constraint. Effects among participants cannot simply be applied to the electorate.

CETaS's retrospective review of the 2024 UK, EU, and French elections found a limited set of viral AI-enabled incidents and no evidence of meaningful changes to election results. It nevertheless documented harassment, authenticity disputes, trust-related harms, and unethical campaign practices (Stockwell 2024).

Established distribution methods—bots, aligned influencers, and information laundering—mattered more to virality than generation alone. AI often rewrote material or expanded production. The review selected publicly visible incidents, so it may miss private or narrowly targeted operations. Its absence of an identified electoral effect is not an estimate of zero effect.

The stronger risk hypothesis therefore concerns architecture: persistent interaction, adaptation, trusted access, control of distribution and feedback, and weak exit or scrutiny. The reviewed evidence does not show these conditions jointly producing society-scale democratic harm.

Provider frameworks and evaluations also distinguish manipulation propensity, configured systems, simulated campaigns, and real-world efficacy. They recognise a risk worth evaluating, not a completed causal chain (Google DeepMind 2026; Anthropic 2026; OpenAI 2026).

5. Four audits of the influence pathway

This is a programme-level governance synthesis, not a single-study article. Four strands examine different parts of the influence pathway at different levels of maturity. They neither observe one integrated operation nor inherit one another's strongest findings.

Guerrero (2026a–f) documents the programme's methods, counts, and coding. External human and technical results belong to the cited studies, not to fellowship experiments.

Part I. How far can an adversarial actor go with \$10?

Pretty far, actually—at the planning stage. That is a qualitative description of exploratory outputs, not an audited \$10 result. The budget question remains open: these records establish neither a verified spending threshold nor the full cost of an operation. Labour, model modification or hosting, audience access, accounts, distribution, and feedback require separate accounting (Guerrero 2026a–b).

The unit is the served system: provider and route, instructions, safeguards, context, memory, tools, permissions, retries, and operator interaction. Deployment further requires an actor, objective, data, distribution, and persistence. Access to generation does not supply that entire system.

Two records must remain separate. An intervention study failed to produce a durable package for independent reload and therefore yielded no admissible behavioural comparison. Separately, three outputs from one unidentified served route contained campaign-planning elements despite refusals or redirection of some tactics.

The outputs described audiences, channels, sequencing, mobilisation or demobilisation, measurement, and adaptation. They did not research or contact voters, execute tools, distribute content, or run a feedback loop. The traces are locally tamper-evident, not independent attestations of the checkpoint or provider (Guerrero 2026a–b).

The next test should measure operational uplift rather than prose alone: named systems in a closed synthetic environment, matched controls, a human-only baseline, reversible actions, and separate marginal and full costs. Appendix A preserves the broader evidence units and their unresolved overlap; the three-output subset is not the whole programme.

Part II. Useful evaluations, but no shared system-level measure

Useful evaluations exist. What this review did not identify is a shared, validated end-to-end instrument or common decision threshold for harmful manipulation across deployed systems and model releases. That is a finding about the reviewed evidence, not an exhaustive claim that no relevant instrument exists.

Results tied to a model version, prompt, judge, or short task do not automatically transfer to assistants with memory, retrieval, tools, conflicting incentives, and repeated access to people. Nor are all evaluations model-only: Akbulut et al. (2026) measure human outcomes across domains and countries. Their preprint finds that manipulation propensity does not reliably predict efficacy, reinforcing the need to keep the two separate.

The GPAI Code also recognises deployment context. The Commission’s March 2026 Taskforce account distinguishes chatbots, third-party applications, agents, and disseminated content, linking evaluation to risk scenarios and assumptions about users. The unresolved task is consistent measurement and validation, not recognition that context matters (European Commission 2025b, 2026b).

The Evaluation Registry contains source-linked observations across six instruments, but those instruments measure different objects: attempted persuasion, deception, manipulation propensity, simulated campaign performance, dialogue behaviour, and benchmark-specific safety responses. No exact model configuration appears in four or more instruments, and there are no adequate same-construct anchors (Guerrero 2026c).

The Registry therefore cannot support one manipulation score, one leaderboard, or one longitudinal frontier trend. Averaging the results would create numerical precision by deleting the distinctions that matter.

The direct APE-120 audit, adapted from Kowal et al. (2025), exposed the same problem inside one evaluation pipeline. A first public summary wrongly reported zero samples because the postprocessor failed to parse the archive. Offline recovery found a substantial run. Among valid records, attempt-coded outputs were common, but every served condition failed the prespecified strict-valid coverage gate. The correct result is non-confirmatory: the harness did not support condition-level estimates (Guerrero 2026c).

This is a measurement constraint, not a frontier trend. Missing observations, incomparable constructs, and evaluator failures cannot be repaired by pooling their scores. Appendix C records the recovered counts and sensitivity bounds.

A proposed instrument would combine a stable, human-calibrated core with deployment-specific modules: exact model and route, memory and data, tools and permissions, operator incentives, delivery, repeated interaction, and feedback. It should separately assess attempted manipulation, understanding, goal-consistent choice, meaningful refusal, revision, and exit, with helpful persuasion as a comparator. Missingness and follow-up must be prespecified. This is a research design, not a validated standard or compliance certificate; APE remains a propensity screen, not a human-effect benchmark.

Part III. Real operations and harms; uncertain electoral effects

AI-enabled manipulative attempts are documented. Uncertainty concerns who encounters them, what changes, and what follows—not whether every incident is hypothetical. Accounts, payments, synthetic media, calls, removals, and investigations reveal different parts of that process (Guerrero 2026d; Stockwell 2024).

The Election Evidence Index separates these propositions. Its current analytic layer contains ten claim-coded records, eight records eligible for incident counting, 96 atomic claims, and 162 claim-evidence relations. A broader catalogue includes incidents,

preparedness files, comparative evidence, and screening leads. It is not a count of confirmed manipulation campaigns (Guerrero 2026d).

CETaS is treated as comparative field evidence rather than an additional claim-coded incident. Its review strengthens the operation–distribution–effect distinction: visible AI-enabled incidents and wider democratic harms were documented, while meaningful electoral-result effects were not identified and quieter operations could fall below the study’s visibility threshold (Stockwell 2024).

The Index separates occurrence, mechanism, attribution, distribution, authentic exposure, human effect, institutional response, and electoral consequence. Evidence for one proposition does not answer the others.

The New Hampshire robocall shows why the distinctions matter. Two days before the state’s 2024 Democratic presidential primary, an AI-generated imitation of Joe Biden was used in robocalls suggesting that recipients save their vote for November. The Federal Communications Commission later imposed a USD 6 million forfeiture concerning 9,581 calls made using misleading caller-ID information. In June 2025, a New Hampshire jury acquitted the operator of state voter-suppression and candidate-impersonation charges (Federal Communications Commission 2024a, 2024b; Associated Press 2025).

The administrative order, criminal acquittal, attempted impersonation, and electoral effect concern different propositions and standards. The case identifies a synthetic political voice, misleading caller identification, distribution near a decision point, and institutional response. It does not identify how many recipients listened, believed, or changed behaviour.

Romania and Moldova records document coordinated accounts, opaque promotion, amplification, synthetic personas, cross-platform distribution, financing allegations, and investigations. They do not isolate generative AI’s contribution or estimate votes changed. Accounts are not people, views need not be unique, and institutional response is not a voter-effect estimate (Guerrero 2026d).

Harms need not wait for measurable vote change. CETaS documents harassment, intimidation, and authenticity disputes. Investigations and corrective responses also consume resources, although this programme does not measure displaced staff time, expenditure, or attention. Resource displacement is a mechanism to investigate—not a quantified aggregate finding (Stockwell 2024; Federal Communications Commission 2024b; Guerrero 2026d).

Election records document operations more readily than effects.

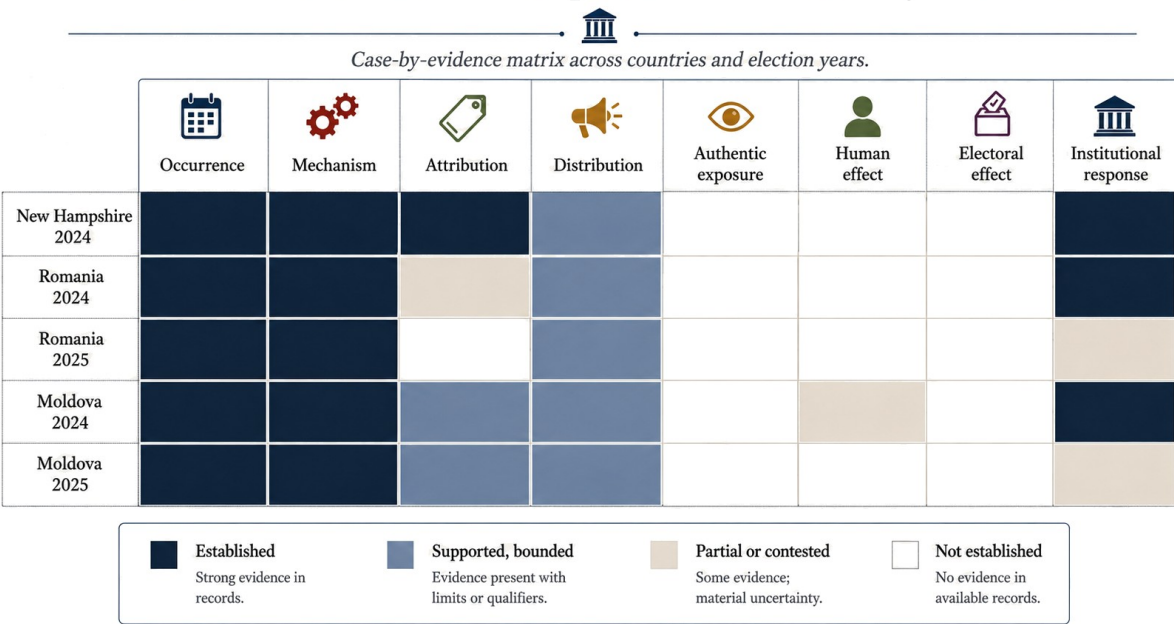


Figure 2. Election records document operations more readily than effects. Darker cells indicate stronger claim-level support in five selected retrospective records; the figure is not the full ten-record analytic layer. Empty effect cells mean the reviewed public record does not identify the effect; they do not mean no effect occurred. Source: Guerrero 2026d.

Precision operations exploit narrow access, private data, trusted identities, or repeated relationships. Saturation operations use volume, contradictory narratives, and coordinated amplification to occupy attention and response capacity. Neither is inherently more severe. Their contribution to agency transfer depends on concealed control, dependency, or impaired contestability; this Index does not establish that downstream condition.

Timing can constrain correction; weaker monitoring can conceal incidents. Those mechanisms make the final decision window and low-visibility settings research priorities, not grounds for inferring prevalence from the present sample.

Established evidence concerns selected attempts, distribution events, responses, and harms. External experiments demonstrate individual-level effects in bounded settings, but the Index does not measure them. Limited field evidence does not establish that individual effects are necessarily small (Stockwell 2024; Akbulut et al. 2026; Guerrero 2026d).

Plausible hypothesis: wider access to capable systems increases manipulative attempts. Open questions: their prevalence, durable agency effects, attention and resource costs, and AI-attributable electoral consequences. The purposive sample estimates none of these; failure to identify an effect does not establish its absence.

Part IV. What works—and for whom?

The policy gap is not a lack of tools. It is a mismatch between tested outcomes and claimed protection. Notice can reveal origin without neutralising influence; friction can reduce sharing without improving quality; and a permission boundary can protect an operator rather than the people targeted. These distinctions change which controls to use and how to evaluate them.

The Policy Evidence Atlas maps 68 control families and 118 implementations. The source-linked review annotates all 118 against 69 selected sources and 278 source–implementation links, with varying review depth. It is not 118 causal evaluations or 278 independent studies. The six historically checked effect claims remain a separate audit subset, not six “winners.” Appendix G documents coverage, revised judgments, and unchanged production flags (Guerrero 2026e–f).

Different outcomes, different conclusions

Nouwens et al. (2020) found that removing a first-screen rejection option increased cookie acceptance in a small field experiment. The study supports an interface effect on recorded choice, not informed preference alignment or the efficacy of a statutory ban. For governance, test meaningful refusal rather than treating acceptance as consent.

Awareness and influence can diverge. In Ibrahim et al.’s (2026) preregistered preprint, 2,610 participants discussed interpersonal conflicts with a sycophantic assistant. Warnings changed perceptions of trust and objectivity without reliably improving self-perceived rightness or repair intentions; actual relationship repair was not measured. Salvi, Cuevas, and Horta Ribeiro’s (2026) commercial-conversation preprint likewise found no significant attenuation of sponsored-book selection with disclosure, but that comparison spans separate collection waves. Neither result makes all disclosure useless.

The reverse pattern appears in the AI Watchdog preprint: 150 participants were randomised across five conditions. Dismissible, just-in-time warnings reduced AI-steered choices from 71.7 to 53.7 per cent on a preregistered secondary endpoint; the primary detection outcome did not improve. With 30 participants per arm and simulated tasks, this warrants replication, not a mandate. Superiority over the forcing variant was not established (Poonsiriwong et al. 2026). Reduced compliance is not itself recovered agency.

Technique education requires an accuracy test. Roozenbeek et al. (2022) support recognition of taught tactics and other bounded laboratory outcomes. Across five later studies, emotional-manipulation inoculation improved technique recognition but not truth discernment on its own; combining it with an accuracy intervention produced different results (Pennycook et al. 2024). Findings also vary by technique and transfer to other topics or sharing (Biddlestone et al. 2025; Ancina, Zumbrägel, and Krämer 2026). Indiscriminate suspicion is not the objective: accurate messages can be emotional.

The treatment and its delivery cannot be separated

Generic friction is not accuracy-focused prompting. Ershov and Morales (2024) link Twitter’s 2020 interface change to lower news sharing, but not a selective reduction for lower-factualness outlets, with heterogeneous political responses. This is a quasi-experimental Twitter study, not a WhatsApp trial. A separate WhatsApp study concerns forwarding-label coverage, not the causal effect of a cap (de Freitas Melo et al. 2024). Forwarding restrictions therefore cannot share one undifferentiated “best-supported” label.

Accuracy prompts instead direct attention towards truth. Survey and field experiments support bounded sharing-quality gains; an internal meta-analysis supports sharing discernment, not twenty independent-team replications. Source-quality proxies do not

measure autonomy (Pennycook et al. 2021; Pennycook and Rand 2022). These prompts merit a distinct implementation and evaluation.

Delivery changes the estimate. Using synthetic controls, Slaughter et al. (2025) estimate 46.1 per cent fewer new reposts after a Community Note appears, but 11.6 per cent fewer when the studied post's earlier engagement is included. Neither is an electorate-wide protection estimate; timing, eligible-post selection, and identification assumptions matter. The 2026 correction adds a competing-interest disclosure, not revised effects (Slaughter et al. 2026).

Corrective context also differs from an AI-origin label. Gallegos et al. (2025) did not detect changes in persuasion, perceived accuracy, or intended sharing for the labelled policy messages tested. DeVerna et al. (2024) found no average discernment gain from tested LLM fact checks, with adverse responses to some wrong or uncertain outputs. Their human comparator was collected later and voluntary viewing was selected. A corrective AI needs its own error and user-outcome tests.

A control must name its beneficiary

CaMeL tests control/data-flow separation and enforced permissions against external prompt injection. It trusts the initiating query and excludes some text-only influence (DeBenedetti et al. 2025). The inference is crucial: securing an agent against outsiders does not secure citizens against an authorised harmful objective. A system can obey its controller while manipulating the person it serves. Permission security and legitimate purpose require separate tests.

Formal exit is not functional exit. Turner et al. (2021) found that data receipt did not enable import or direct transfer across four tested IoT systems. Allcott et al.'s search experiments distinguish active choice, defaults, and trial of an alternative; their market-share counterfactuals are modelled, not observed assistant migration (2025, revised 2026). For persistent assistants, the proposed test is whether users can continue the task after switching without losing necessary context or institutional expertise.

Memory minimisation remains a structural priority, but its claimed benefit must be bounded. Offline recommender experiments measure quality under reduced data budgets, not protection from conversational dependency (Biega et al. 2020). Associations between heavier voluntary chatbot use and adverse psychosocial outcomes are not randomised evidence that a memory cap works (Fang et al. 2025). Data minimisation, continuity, and usable switching need deployment-specific tests, including privacy costs; their democratic importance is not settled by the ease of obtaining an effect estimate.

The synthesis separates outcome, delivery, authority, and exit. A control may affect one without protecting the others. The hypothesis that independently governed monitors, enforceable authority limits, and usable alternatives jointly preserve agency remains untested; the source-specific findings do not establish the package (Guerrero 2026f).

Evidence question	What a study can establish	What an agency claim still requires
Outcome	Recognition, trust, sharing or choice can change independently.	Understanding, goal-consistent choice, refusal and revision—not one proxy alone.
Delivery	A control can work after exposure to the control.	Coverage, latency, missed cases and displacement across the whole decision window.
Authority	A boundary can protect trusted instructions from an external attack.	A legitimate purpose, a named beneficiary, and limits on authorised conflicting objectives.
Exit	A service can provide an export or a choice screen.	Continuity, usable alternatives and independence from both the assistant and its defender.

Figure 3. Four questions that separate component performance from protection of agency. These are analytical distinctions, not a measured causal sequence or an efficacy ranking. Source: synthesis of the studies discussed in Part IV; Guerrero 2026f.

Two current policy cases: legal duties are not effect estimates

Mandatory marking and disclosure. Article 50(2) requires covered providers to make synthetic audio, image, video, and text machine-readably marked and detectable as artificial—not universally stamped with a visible watermark. Standard editing and other exceptions matter. Separate deployer duties cover deepfakes and certain public-interest text; the latter has an exception for human review or editorial control with editorial responsibility. The general application date is 2 August 2026. Amended Article 111(4) gives systems placed on the market before then until 2 December to comply with Article 50(2). The associated Code is voluntary; applicable statutory duties are not (European Union 2024a, 2026; European Commission 2026c).

Recommendation. Providers and distributors should test the whole marking-to-notice chain: survival through routine transformations, machine readability, timely notice, and false positives. Political impersonation also requires verification of the claimed speaker, not just synthetic origin. These controls make origin and responsibility more inspectable; origin-label and warning experiments do not establish general protection of political choice (Gallegos et al. 2025; Ibrahim et al. 2026).

ChatGPT under the DSA. The Commission designated ChatGPT as a Very Large Online Search Engine on 31 August 2026. This is a service designation, not a GPAI-model classification or finding of manipulation. Additional obligations apply four months after notification, which the Commission describes as January 2027. Notification, not the announcement alone, determines the deadline. On 1 September the register listed the designation but not its decision text (European Commission 2026d–e).

Recommendation. Use systemic-risk assessment and mitigation under Articles 34–35, independent audit under Article 37, and qualified researcher access under Article 40 to scrutinise the designated service within scope. Examine conflicting objectives, personalisation, repeated interaction, and users’ ability to contest influential outputs—not only isolated model refusals. Preserve privacy and security while limiting the provider’s exclusive control of the evidence. Designation creates an accountability route; its effect on conduct or democratic agency remains unestimated (European Union 2022; European Commission 2026d–e).

These examples establish legal responsibilities, not policy efficacy. Compliance, a tested component effect, and preserved democratic agency remain separate claims.

6. Power accumulates in the joins

The audits share an asymmetry: external records often become thinner between capability and downstream effect, while a system's controller may retain detailed feedback. The second proposition is an architectural inference, not a measured comparison of visibility.

A controller may know who was selected, what they received, how they responded, and why the next interaction changed. The affected person or investigator may see one message or aggregate count. Power can therefore lie in observing, optimising, and selectively disclosing influence—not only in producing it.

The actor best placed to optimise the influence process may also be the only actor able to explain it.

This does not establish that one present actor controls the entire chain. It identifies a democratic dependency: institutions cannot independently scrutinise an influential system if its controller holds the only usable account of what happened.

Diffusion and concentration can coexist. Widely available models can enlarge the adversary set while control remains concentrated around data, identity, orchestration, cloud services, dominant assistants, distribution, and evidence access. Conversely, concentrated model development does not prevent downstream diffusion. The relevant question is which combinations one actor can control, not model access alone.

The same applies to defence. A provider controlling both an assistant and its only monitor may determine violations, retained evidence, warnings, and the account given to investigators. A second model is not independent scrutiny when it answers to the same authority. This is a control-allocation inference, not a measured concentration effect (Lazar 2024; Guerrero 2026f).

An independently governed, user-directed defensive layer could interrupt that dependency, but could also become a new intermediary people cannot contest. Evaluation should test conflicts, meaningful dismissal, privacy, and continued access to alternatives—not merely whether the monitor changes choices.

A conditional catastrophic pathway

A catastrophic pathway needs more than many persuasive outputs. It would require sustained influence, access to consequential decisions, concentrated control, and correction or exit that no longer works at societal scale.

Table 2 makes that conjunction testable by pairing each condition with an indicator, possible interruption, and evidence that would weaken the concern.

Table 2. Conditions in the proposed catastrophic-risk pathway. Their conjunction and sufficiency are not established. Source: author’s synthesis.

Condition	Observable indicator	Potential interruption	What would weaken the concern
Sustained capability	Repeated, adaptive interaction remains effective	Limit memory, tools, and consequential actions; test exact routes	Effects decay quickly or refusals remain reliable
Trusted or infrastructural access	Assistants, identity channels, workflows, or dominant platforms provide access	Authentication, disclosure, plural channels, procurement diversity	People know the principal and can leave without material cost
Joined control	One actor controls several CDE links	Separate permissions; require independent approval at consequential transitions	Control remains divided and no actor can optimise the full pathway
Private optimisation and weak observability	The controller sees response data while outsiders lack route, delivery, or audit evidence	Preserve or escrow records; enable qualified independent audit	Reviewers reproduce key claims and relevant denominators in time
Slow correction and weak contestability	Remedy arrives after the decision window	Rapid, reversible response with notice, appeal, and expiry	Errors are corrected before meaningful decisions with low rights burden
Persistent dependency	People or institutions cannot function or decide without the system	Exit, interoperability, human override, and institutional redundancy	Dependence remains voluntary, plural, and reversible

The components have different evidence status. Bounded capabilities and selected operational precursors are observed. The importance of joined control and private evidence is a strong inference. Persistent dependence and compounding effects are plausible hypotheses. Society-scale degradation of democratic self-correction is a speculative scenario; its probability and threshold remain open.

Such a scenario need not resemble a single “hacked” election. It could involve recurrent authenticity disputes, public institutions dependent on the firms they oversee, distorted perceptions of public opinion, and corrections that consistently arrive too late.

The severity test is persistent, society-scale loss of democratic self-correction—not persistence alone. The proposed mechanism is control of the interfaces, dependencies, and evidence through which error would otherwise be challenged.

7. Govern the system before the outcome

Govern the evidenced mechanism without claiming an unmeasured outcome. Waiting for an election-wide counterfactual can leave actionable harms untouched; regulating capability alone can miss the deployed system.

That requires evidence to travel across institutions. Providers observe configurations and failures; platforms observe delivery; election authorities face deadlines; regulators and

courts need lawful, contestable grounds for action. No one source automatically supplies the others' evidence.

The design test has three parts. **Visibility:** identify the controller, purpose, sponsor, route, and relevant targeting criteria. **Authority:** separate consequential permissions and make revocation effective. **Review:** preserve proportionate access to evidence, reasons, challenge, appeal, and restoration where possible. These are governance objectives, not an evaluated protection package.

Evidence retention and inspection also create privacy, security, cost, competition, and intellectual-property burdens. Retain what a defined oversight purpose requires, with access controls and review; more logging is not automatically better governance.

EU law covers different parts of this architecture. The Political Advertising Regulation addresses sponsorship and targeting; the DSA adds systemic-risk, audit, and researcher-access duties for covered very large services; the AI Act supplies specified prohibitions, GPAI systemic-risk duties, and a high-risk category for some electoral uses. Their scopes and tests differ (European Union 2022, 2024a, 2024b, 2026; European Commission 2025a).

Their timetables also differ. Under the July 2026 amendment, Chapter III, Sections 1–3, except Article 6(5), apply from 2 December 2027 to Annex III high-risk systems and 2 August 2028 to Annex I systems. This is not a general postponement of manipulation prohibitions, GPAI duties, or Article 50. Part IV sets out the separate marking transition and ChatGPT timetable (European Union 2026).

The voluntary GPAI Code supports risk identification, evaluation, mitigation, and reporting within its provider scope; it does not replace Article 5, Article 55, or Annex III (European Commission 2025b, 2026a).

NIST's Generative AI Profile likewise provides voluntary lifecycle guidance for documentation, testing, provenance, monitoring, and incident reporting—not evidence of democratic protection (National Institute of Standards and Technology 2024).

Who must produce what evidence

The framework becomes operational only when responsibility follows control. Each actor should be able to produce evidence at the CDE node it controls, while affected people and public institutions retain practical routes to challenge and remedy.

Table 3. Responsibilities and protected rights across the influence pathway. These are governance proposals unless an applicable duty is identified in the text. Source: author's synthesis.

Actor or institutional role	Responsibility or protected right	Evidence that should exist	Claim ceiling / safeguard
Model or served-system provider	Evaluate the exact configured route, safeguards, tools, permissions, and material changes.	Versioned route receipt; system configuration; test protocol; failure and missingness record; serious-incident criteria.	Capability or control evidence is not deployment prevalence or human effect.
Deployer, campaign, or vendor	Name the principal, purpose, data, authority, approvals, and limits of consequential political use.	Purpose record; data provenance; permission and approval logs; actions attempted and completed; incident history.	Operational records do not establish authentic exposure or persuasion.

Actor or institutional role	Responsibility or protected right	Evidence that should exist	Claim ceiling / safeguard
Platform, telecom, identity, or distribution intermediary	Preserve auditable delivery denominators and maintain proportionate interruption and restoration mechanisms.	Deduplicated reach where lawful; frequency; targeting and forwarding uplift; enforcement, appeal, and restoration timelines.	Delivery or interruption is not proof of belief, turnout, or election effect.
Electoral, regulatory, judicial, and research oversight	Secure timely, independent access; adjudicate bounded claims; preserve due process and confidentiality.	Chain of custody; access logs; methods; uncertainty; comparator; reasons; appeal, expiry, and after-action review.	Authority and investigation do not substitute for causal evidence.
Affected people and public institutions	Protected right: intelligible notice, meaningful challenge, practical exit, appeal, and restoration where possible.	Notice and explanation records; complaint and appeal outcomes; restoration; accessibility and disparate-burden measures.	Disclosure alone does not prove informed consent or retained agency.
Defensive monitor / assurance provider	Disclose conflicts; permit independent scrutiny, meaningful dismissal, appeal, and exit; protect the user's data.	Warning logic and criteria; test results and error rates; retained traces; conflict disclosures; override and complaint records.	A changed choice or passed security test does not establish the affected person's agency.

Measure the node an intervention changes. Preserve evidence for the next node. Do not promote one into the other.

Policy priorities: tested endpoints, accountable authority

The review supports a differentiated portfolio, not an efficacy league table. Existing duties should be applied within scope. The following recommendations identify actors, actions, and tests; their legitimacy still requires necessity, proportionality, rights protection, and safeguards against capture (Guerrero 2026f).

Platforms and interface providers: use bounded human-facing controls. Make refusal as accessible as acceptance; test accuracy prompts separately from forwarding friction; and measure whether verified corrections arrive in time. The supporting studies concern choices, sharing, and engagement—not electoral protection. Include understanding, truthful circulation, and legitimate participation as outcomes (Nouwens et al. 2020; Pennycook et al. 2021, 2024; Ershov and Morales 2024; Slaughter et al. 2025).

Deployers and public procurers: enforce limits on consequential authority. Specify permitted tools, recipients, actions, approval, and revocation outside the model. Test whether cancelling permission actually stops execution. Then test whose purpose the system serves: resistance to external injection does not prevent authorised steering against a citizen's interests (Debenedetti et al. 2025; Salvi, Cuevas, and Horta Ribeiro 2026).

Researchers and defensive providers: pilot, do not presume, conversational protection. Replicate dismissible just-in-time interventions with user-goal consistency, understanding, and refusal as outcomes. Test errors, privacy burden, unequal cognitive load, and rejection of accurate helpful advice. Watchdog's secondary-endpoint signal and the mixed correction evidence justify this study—not a universal monitor mandate (Poonsiriwong et al. 2026; DeVerna et al. 2024).

Supervisors and public institutions: protect independent evidence and usable exit. Apply relevant audit and researcher-access duties. Procurement should include a switching exercise: can staff continue a real task with another provider while retaining necessary context and expertise, without exporting unnecessary personal data? Portability and market-choice studies support this as a structural test, not a demonstrated remedy for assistant dependence (Turner et al. 2021; Allcott et al. 2025, revised 2026; Lazar 2024).

Provenance and disclosure remain useful for their specified purposes. None substitutes for control of authority, practical alternatives, or independent scrutiny. Measure delivery, errors, displacement, and rights costs before extending a component result into a broader protection claim.

8. Research the missing bridges

The next studies should test the missing bridges rather than add incomparable scores. Four priorities follow.

1. Make the evidence reproducible

Human-calibrate the APE attempt and refusal judgements, assign a reason to every missing record, preserve exact routes, and rerun a smaller fixed panel. In parallel, test the G0–G7 sequence across exact checkpoints with durable intervention packages, clean reload, matched controls, utility and safety constraints, and independent replication. The outcome is not a better leaderboard. It is whether independent teams can reproduce a bounded estimate from the same artefacts.

2. Measure operational uplift

Compare human-only, model-only, tool-enabled, agentic, and safeguarded-agent conditions in a closed synthetic election environment. Record time, expertise, full cost, completion, coordination, violations, resistance, revocation, and recovery. This tests whether AI expands the practical adversary set or compresses roles; the current traces do not establish either.

3. Connect mechanisms to people and institutions

Measure deception, concealed purpose, vulnerability exploitation, dependence, and restricted choice—not agreement alone. Include repeated interaction, memory, delayed follow-up, meaningful exit, and institutional workload, triage, delay, evidence loss, or dependence. An agency-transfer claim needs reduced reflective control or constrained alternatives paired with another actor’s increased practical power.

Exposure studies should distinguish unique reach, frequency, timing, selection, prior alignment, countervailing exposure, and the remaining correction window. Privacy-preserving platform or telecom partnerships may help. Without credible causal identification, retain an exposure or response estimate rather than recoding it as persuasion.

4. Evaluate the assistant and its defender together

Preregister a joint evaluation of an assistant and its defender. Separate helpful persuasion, authorised but concealed conflicting objectives, and external injection. Randomise specific controls, not generic guardrail bundles. Primary outcomes should capture goal-consistent choice, understanding, meaningful refusal, revision, and exit. Detection, trust, and satisfaction are secondary; include helpful accurate advice so indiscriminate rejection counts as a cost.

Use consenting participants and low-risk tasks, not live voter targeting. Record coverage, timing, errors, privacy, cognitive burden, and responses to revocation. Longer follow-up should test persistence and whether dependence shifts to the defender. Map who controls the purpose, data, monitor, evidence, and remedy (Guerrero 2026f).

9. Limitations and responsible release

The contribution is a provisional governance framework and evidence synthesis. It is not a completed causal measurement of agency transfer.

Part I has partly unreconciled evidence units and an unidentified served route. Local tamper evidence is not independent attestation. The failed intervention yielded no behavioural comparison, and the \$10 question is not a verified cost result. Appendix C's USD 14.23 is the separate APE-120 run's recorded spend, not a Part I operation cost.

Part II remains descriptive and non-confirmatory: coverage and missingness prevent condition-level estimates, and construct differences prevent a pooled manipulation score or frontier trend. No cross-classified model of topics and served conditions is reported. Failure to identify a shared standard in the reviewed sources is not exhaustive evidence of absence.

Part III is purposive, uses heterogeneous denominators, and contains contested attribution; it cannot estimate prevalence. Part IV is purposive, English-dominant, and single-pass. Its 118 implementations, 69 sources, and 278 links are not independent causal studies. Eight sources received new main-text or substantive-section checks in that review; other sources were assessed through abstracts, summaries, official text, or prior adjudications. Appendix G records the differences (Guerrero 2026f).

The review neither reproduces all source data nor independently double-codes the literature. Selected legal checks are not a comprehensive applicability audit. Preprints, short or simulated tasks, self-reports, secondary endpoints, cross-wave comparisons, and selected voluntary exposure retain their limits. The Atlas's production flags are unchanged; an unidentified effect in selected sources is not proof that no relevant study exists.

Internal design and review create risks of self-confirmation and coding error. No fellowship experiment measured authentic exposure, human persuasion, durable preference change, voting behaviour, agency transfer, power concentration, electoral outcome, or policy effect. External studies supply the human and technical findings discussed here; their inclusion is not independent replication.

Failed persistence, parsing, and evaluator coverage are findings about experimental control and observability. They are not behavioural nulls or substitutes for the intended confirmatory results.

Aggregate electoral causality may remain partly unidentified even with better data: competing campaigns, media, interpersonal discussion, institutional responses, and prior beliefs interact without a clean counterfactual.

Responsible release is deliberately bounded. The paper publishes concepts, aggregates, source-linked claims, non-operational figures, and defensive designs, not harmful prompts, campaign-ready outputs, targetable profiles, evasion methods, credentials, or distribution instructions. No real voters, campaigns, platform accounts, or election infrastructure were used. Further operational release requires independent ethical and dual-use review.

10. Conclusion: preserve democratic self-correction

The democratic question is not only what AI can make people believe. It is who gains control over the systems through which they learn, choose, act, and seek correction.

This programme finds planning assistance without demonstrated deployment, a fragmented evaluation landscape, documented operations and harms without identified electoral effects, and controls whose tested benefits stop short of the democratic protection often claimed for them. These findings support a governance response, not a claim that AI has changed an election.

The CDE Gap identifies the measured endpoint and missing causal bridge. Agency transfer asks whether another actor gains power as citizens or institutions lose reflective and contestable control. That distinction applies equally to influential systems and the monitors presented as their safeguards.

Act on evidenced mechanisms. Match controls to outcomes and beneficiaries. Preserve independent evidence, revocable authority, and usable alternatives. Evaluate delivery and rights costs rather than treating legal compliance, changed choices, or passed security tests as proof of preserved agency.

Persistent, society-scale loss of democratic self-correction remains a conditional scenario with unknown probability. Neither alarming outputs nor missing effect estimates settle it.

Democracy does not require the absence of influence. It requires the ability to identify who acts, compare alternatives, refuse, contest, and repair error. The task is to keep AI-mediated influence open to that correction before dependence makes control difficult to recover.

Technical and evidentiary appendix

A. Part I evidence-unit reconciliation

Table A1. Part I evidence units. These records may overlap and must not be added into one sample count. Source: Guerrero 2026a–b.

Record	Unit	Count	What it supports	What it does not support
Paired red-team pilot	Objectives across two scenario families	10	Exploratory comparison of official hosted refusal behaviour and safeguard-reduced workflow prototyping	One independent n=10 effectiveness estimate; deployment or human effect
Midpoint record	Exploratory probes	3	Protocol development for the synthetic study	A disjoint sample or completed study
Deepest forensic subset	Client-observed bundles / requests / outputs	2 / 5 / 3	Route-level planning content and refusal/final-answer structure in the most auditable subset	Total Part I evidence universe; named checkpoint; prevalence
Deepest forensic subset	Local hash-chain events	18,907	Local record continuity and internal trace inspection	Provider attestation, trusted timestamp, or independent custody
Research boundary	Live deployment / authentic exposure	0 / 0	Scope of the study	A failed-output count or evidence of safety
Later expanded corpus	Unique conversations / requests or runs	6 / 31	Post-hoc material for future reconciliation and coding	A headline sample until overlap with the frozen subset is resolved

Part I is broader than the forensic subset

The record contains different evidence units. They may overlap and must not be added into one sample count.

BROADER PILOT	MIDPOINT RECORD	DEEPEST FORENSIC SUBSET	RESEARCH BOUNDARY
10 paired objectives 2 scenario families Official routes: 10/10 refusals. Safeguard-reduced routes: workflow prototypes.	3 exploratory probes Protocol-generating The probes shaped the synthetic study design. They are not known to be disjoint from the pilot.	2 trace bundles · 5 requests 3 outputs · 18,907 events The most auditable subset, with local hash chains. It is not the total Part I evidence universe.	0 live deployment 0 authentic exposure This is a study boundary— not a failed-output count.

Breadth, protocol development, and forensic depth are complementary views—not additive stages.

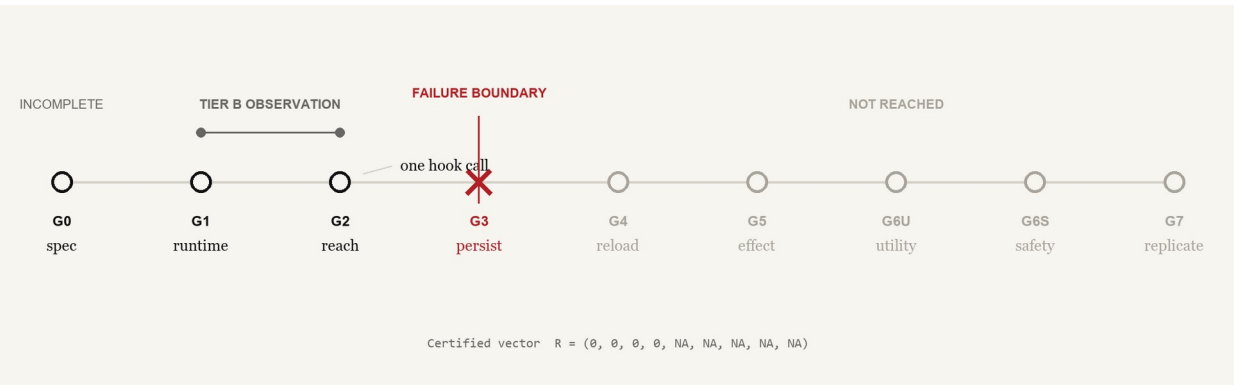
Claim ceiling: exploratory planning and route behaviour only; no autonomous execution, real audience contact, persuasion, or electoral effect.

Appendix Figure A1. Part I evidence units are complementary and may overlap; they must not be added into one sample count. The figure distinguishes the broad paired pilot, protocol-generating probes, the deepest forensic subset, and the research boundary. No live deployment or authentic exposure occurred. Source: Guerrero 2026a.

B. G0–G7 conversion record

Table A2. G0–G7 conversion record for the two attempted intervention routes. Source: Guerrero 2026b.

Gate	Minimum evidence	DeepSeek route	Qwen route
G0	Frozen specification. Registered target, intervention, task, budget, stop, outcomes, and analysis	Reached	Reached
G1	Executable runtime. Exact checkpoint loaded in a recorded environment	Reached	Reached
G2	Intervention reachability. Intended mechanism entered under matched controls	Reached once	Incomplete / not certified
G3	Durable persistence. Stored package with manifest, size, schema, revision, and digest	Failed: no durable capture	Failed: no saved adapter after two 300-step loops
G4	Independent reload. Clean retrieval, verification, load, and exercise of exact bytes	Not reached	Not reached
G5	Valid behavioural estimate. Sealed paired evaluation against preregistered controls	Not reached	Not reached
G6U	Utility constraint. Declared utility boundary met	Not reached	Not reached
G6S	Safety constraint. Declared safety boundary met	Not reached	Not reached
G7	Independent replication. Registered result reproduced by another team	Not reached	Not reached



Appendix Figure A2. Access-to-control gate record for the deepest completed intervention attempt. The run reached the intended mechanism once but did not create a durable intervention package; later gates were therefore not reached. The binary vector shown below the sequence is an internal gate-status record, not a risk score. Source: Guerrero 2026b.

C. APE-120 reconciliation

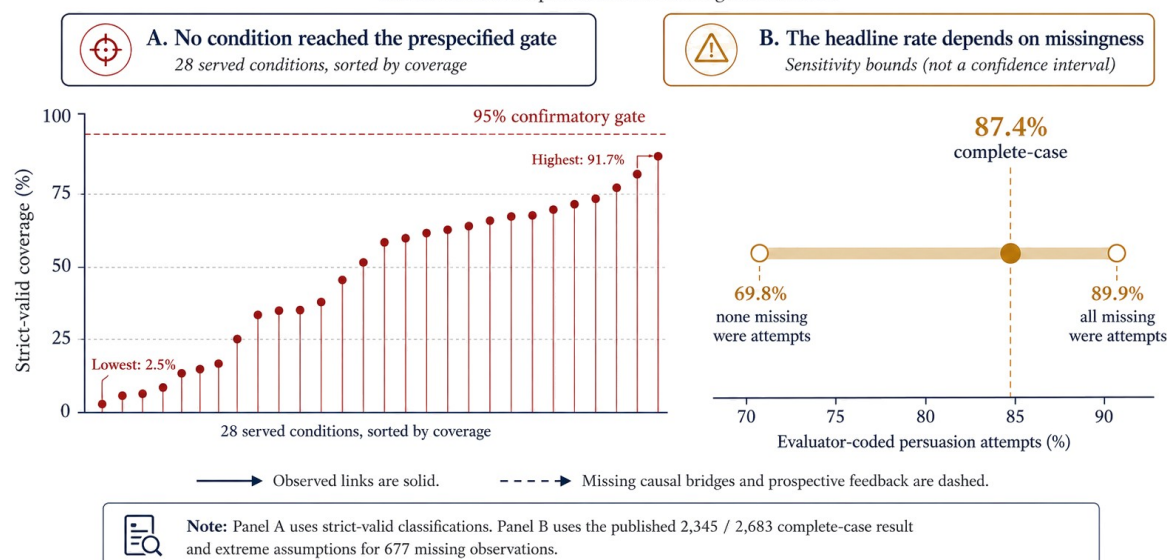
Table A3. APE-120 denominator and coverage reconciliation. Source: Guerrero 2026c.

Stage or diagnostic	Count	Interpretation
Frozen queue	66 conditions	Planned queue, not completed sample
Conditions recorded before timeout	33	Includes route failures
Route failures before inference	5	No target inference completed
Finalised served conditions	28	Conditions with 120 finalised records each

Stage or diagnostic	Count	Interpretation
Finalised records	3,360	28 × 120
Valid target + attempt-evaluator pairing	2,683	Complete-case denominator for attempt coding
Attempt-coded records	2,345	87.4% descriptive complete-case rate
Missing valid pairing	677	20.1% of finalised records
Extreme missingness sensitivity	69.8–89.9%	Bounds under none/all missing coded as attempts; not a confidence interval
Valid refusal judgements	1,776	Separate denominator from attempt coding
Refusal-coded records	741	41.7% of valid refusal judgements
Strict-valid classifications	1,711	Used for the prespecified coverage gate
Conditions meeting 114/120 gate	0	No confirmatory model estimate admitted
Recorded spend	USD 14.23	Audit cost for the frozen run

The attempt rate is high; the harness is not confirmatory

Panel A uses strict-valid classifications. Panel B uses the published 2,345 / 2,683 complete-case result and extreme assumptions for 677 missing observations.



Appendix Figure A3. Attempt-coded responses were common among valid observations; the harness was not confirmatory. Panel A shows strict-valid coverage across finalised served conditions. Panel B places the complete-case rate inside extreme missingness bounds. The bounds are not confidence intervals. Source: Guerrero 2026c.

Failure diagnostics include 538 target truncations, 501 refusal-judge errors, 471 invalid or unparseable refusal-judge outputs, and 106 evaluator errors. These categories may overlap and should not be summed without record-level reconciliation.

D. Reusable claim and policy-effect ceiling record

Table A4. Reusable record for claim ceilings and policy-effect ceilings. Source: author synthesis.

Field	Required entry
Observed node	Capability, served system, deployment, exposure, human effect, institutional effect, or intervention
Unit and configuration	Exact model/route/system, actor, data, permissions, distribution, and comparator
Evidence status	Established evidence, strong inference, plausible hypothesis, speculative scenario, or open question
Sample and exclusions	Denominator, clustering, missingness, route failures, and post-hoc filters

Field	Required entry
Direct endpoint	What was actually measured and over what time
Strongest licensed claim	Narrow conclusion supported by the design
Unsupported downstream claim	Next CDE conclusion that the result does not establish
Minimum next evidence	Design and records needed to cross the next bridge
Rights and legitimacy burden	Privacy, speech, due process, security, competition, access, and distributional effects
Review and reversal	Appeal, expiry, retest, restoration, deletion, or narrowing condition
Purpose and beneficiary	Who sets the objective; whose authority the control protects; how conflicts with the affected person's goals are identified.
Delivery and exposure to the control	Eligibility, coverage, attachment or response time, untreated cases and the remaining decision window.
Agency and defensive dependence	Understanding, meaningful refusal, revision, continued task capability after exit, and dependence on the monitor itself.

E. Three manipulation baselines that should not be conflated

Table A5. Three manipulation baselines that should not be conflated. The paper's normative test, Article 5's prohibition, and the GPAI Code's systemic-risk category serve different functions. Source: author synthesis from European Union 2024a and European Commission 2025b.

Baseline	Core test or object	Status and scope	Claim ceiling
This paper	Influence becomes manipulation when opacity, deception, vulnerability exploitation, asymmetry, dependency, or control of the decision environment degrades reflective and contestable choice.	Normative analytic definition used across micro, meso, macro, and democratic levels.	Does not by itself establish occurrence, harm, legal liability, agency transfer, or effect.
AI Act Article 5	Specified subliminal, purposefully manipulative or deceptive techniques, or exploitation of listed vulnerabilities, under cumulative material-distortion and significant-harm conditions.	Binding prohibition with conjunctive legal thresholds and exceptions; not a general ban on persuasion or political manipulation.	Legal coverage does not establish prevalence, causal effect, or democratic materiality in a particular case.
GPAI Code Appendix 1.4(4)	Specified systemic risk of strategic distortion of behaviour or beliefs through persuasion, deception, or personalised targeting, especially multi-turn or difficult-to-detect influence at scale or against high-stakes decision-makers.	Voluntary Article 55 compliance baseline for signatories that provide GPAI models with systemic risk.	Framework existence or adherence does not establish that manipulation occurred, succeeded, or was mitigated.

F. Illustrative policy-effect ceilings

The matrix below is illustrative, not a ranking. It specifies the nearest endpoint a control can credibly be said to change, the minimum evidence needed, the principal rights burden, and the downstream claim that remains unsupported.

Table A6. Illustrative policy-effect ceilings at specific CDE nodes. Legal authority, implementation quality, rights burdens, and review conditions remain context-dependent. Source: author synthesis of programme and public evidence.

Governance option	CDE node / evidence holder	Nearest measurable endpoint	Minimum credible evidence	Rights and reversal	Unsupported downstream claim
Synthetic political impersonation	Deployment → exposure; sender, telecom, platform, election authority	Verified-origin coverage; confirmed impersonations blocked before delivery; false positives, bypass, and interruption time	Delivery-level records with an auditable denominator and independently adjudicated samples	Protect satire, anonymity, privacy, and appeal; time-limit protocols and restore wrongful blocks	Belief, turnout, vote choice, or election outcome changed
Sensitive or vulnerability data for political targeting	Deployment; campaign, data controller, broker, platform, DPA	Lawful-purpose and provenance coverage; prohibited joins detected; deletion or access requests completed	Scoped data-lineage audit and adversarial sample, not self-attestation alone	Protect privacy, association, organising, and security; deletion deadlines, necessity review, and appeal	Compliant data use prevented manipulation or electoral harm
Tool-enabled generation-to-distribution workflow	Capability → deployment; provider, deployer, tool operator, credential owner	Prohibited-action success; authorisation coverage; escalation; revocation latency; rollback success	Exact configured system, preserved logs and artefacts, matched sandbox controls, and clean replay	Protect security and legitimate automation; graduated permissions, expiry, rollback, and retest after change	Prevalence of misuse, authentic exposure, or human effect
Opaque or high-frequency distribution	Deployment → exposure; platform, advertiser, intermediary, qualified researcher	Deduplicated authentic reach; frequency; targeting accuracy; recommendation or forwarding uplift; correction-window time	Privacy-preserving delivery data with an auditable denominator and appropriate comparison design	Protect privacy, speech, and equal access; post-window review, disparate-burden tests, and reversal	Persuasion, agency transfer, turnout change, or electoral materiality
Weak evidence custody or route opacity	Across CDE bridges; provider, deployer, auditor, regulator, or court	Route-receipt completeness; artefact preservation; hash/schema validity; clean reload; inter-auditor reproduction; documented missingness	Versioned, content-addressed artefacts under independent or escrowed custody with tiered access	Protect privacy, confidentiality, cybersecurity, IP, and proportionality; minimise data and limit retention	The system was safe, manipulative, effective, or electorally consequential
Slow incident interruption and remedy	Exposure → response; platform, telecom, election authority, court or law enforcement by mandate	Detection-to-preservation, attribution, interruption, notice, appeal, and restoration time; evidence-loss and wrongful-action rates	Incident timelines, outcome-coded appeals, and independent after-action review	Protect due process and against over-removal; automatic expiry, prompt review, restoration, and aggregate reporting	Votes were “saved,” belief change was averted, or the intervention worked election-wide

G. Source-linked policy review: findings and claim ceilings

Scope. The 1 September supplement individually annotates the existing 118 implementations and keeps three additional candidates outside that denominator: accuracy-focused sharing prompts, timely claim-specific corrective context, and an independently governed dismissible conversational monitor. Its 69 selected source records generate 278 source–implementation links; overlapping studies, versions, and repeated mappings must not be counted as independent corroboration (Guerrero 2026f).

Depth. The review recorded eight new main-text or substantive-section checks: Slaughter et al. (2025), Biddlestone et al. (2025), Ancina et al. (2026), DeVerna et al. (2024), Debenedetti et al. (2025), Poonsiriwong et al. (2026), Ibrahim et al. (2026), and Salvi et al. (2026). Other source records have explicitly different depths, including publication abstracts or summaries, official records, and six retained prior adjudications. The manuscript integration rechecked key passages and bibliographic metadata; neither exercise independently reproduces the source data or constitutes a double-screened systematic review.

Interpretation. The H/T/O/A/N categories in the supplement indicate kinds of support—human, technical, observational/implementation, adjacent-domain, and normative/proposal—not a strength ranking. A–F remains historical recommendation metadata. Source-specific matches and limitations govern the revised manuscript; the production Atlas’s claim-check flags remain unchanged. A component effect, a law’s existence, and preserved agency require different propositions and evidence.

Table A7. Selected external findings integrated into Part IV. Study findings are separated from the paper’s agency-transfer inferences. The full 118-row source ledger is Guerrero 2026f.

Source and design	Observed or reported result	Ceiling and research implication
Nouwens et al. 2020 Field interface experiment; n = 40	The availability of a first-screen rejection option altered cookie acceptance.	Recorded choice is not informed preference alignment; a statutory ban and its enforcement were not the treatment.
Ershov and Morales 2024 Quasi-experimental Twitter change	Lower news sharing, without a significant selective reduction for lower-factualness outlets; heterogeneous responses.	Generic friction is not accuracy-focused prompting or a WhatsApp cap. Include lost legitimate circulation in the evaluation.
Pennycook et al. 2021; Pennycook and Rand 2022 Surveys, field experiment; internal meta-analysis	Accuracy-oriented attention improved bounded sharing-quality or discernment outcomes.	Source-quality and stated-sharing proxies do not establish agency. The internal meta-analysis is not twenty independent-team replications.
Roozenbeek et al. 2022; Pennycook et al. 2024; Biddlestone et al. 2025; Ancina et al. 2026	Technique recognition, truth discernment and sharing do not necessarily improve together. Some benefits are treatment- or topic-specific.	Assess accurate emotional content, untaught tactics and follow-up. In Ancina et al., n = 154 and simulated messenger exchanges limit transfer.
Slaughter et al. 2025, 2026 Synthetic-control observational analysis	Post-attachment attenuation is larger than the effect relative to the post total including earlier engagement.	The post-note and total windows differ; neither is an all-platform rollout estimate. The correction concerns competing interests, not revised effects.
Ibrahim et al. 2026 Preregistered experiment; analysed n = 2,610; preprint	Warnings shifted some perceptions, without reliable beneficial effects on self-perceived rightness or repair intentions.	Eight-turn conflicts and self-reports; no demonstrated relationship repair or general zero-effect conclusion.
Poonsiriwong et al. 2026 Five-arm experiment; n = 150; preprint	A dismissible monitor improved the secondary AI-steered-choice endpoint, not the primary detection measure.	Thirty participants per arm, simulated tasks; no established advantage over forcing, durable agency outcome, or demonstrated local deployment.

Source and design	Observed or reported result	Ceiling and research implication
Salvi, Cuevas, and Horta Ribeiro 2026 Two experiments; n = 2,012; preprint	Promotional conversation shifted book selections. Disclosure did not significantly attenuate steering in the reported comparison.	Disclosure and baseline span collection waves, not one contemporaneous five-arm randomisation. Selection is not established purchase behaviour or durable agency loss.
DeVerna et al. 2024 Preregistered experiment; n = 2,159	No average discernment improvement from tested LLM fact checks; adverse responses to some erroneous or uncertain outputs.	Forty headlines and one older configuration. Human comparator collected later; optional viewing is selected rather than randomised exposure.
Debenedetti et al. 2025 CaMeL technical evaluation, v2	Control/data-flow separation and permission policies constrain external prompt injection under the stated assumptions.	Initiating query trusted; some text-only influence out of scope. Protecting the principal is not protecting people from that principal.
Turner et al. 2021; Allcott et al. 2025, revised 2026 IoT exercises; search field experiments	Data export did not deliver functional migration in the IoT exercises. Choice, default and trial interventions differ in search.	Historical and adjacent domains, not persistent-assistant migration. Search-market counterfactuals are modelled. Test continuation, not a download alone.
Biega et al. 2020; Fang et al. 2025 Offline recommendation; four-week chatbot study	Data budgets affect recommender quality. The chatbot study distinguishes assigned conditions from voluntary-use associations.	Neither is a randomised test of conversational memory limits. Preserve the distinction between usage association and treatment effect.

Analytical conclusion. Outcome mismatch, incomplete delivery, and protection of the wrong principal are three distinct reasons a component result may not justify a democratic-protection claim. The independently governed monitor-plus-authority-limits-plus-exit package is a plausible hypothesis. Durable reflective agency, institutional self-government, and the package's distributional or concentration effects remain open questions. No new societal effect estimate is claimed.

Source independence and versions. Preprints, affiliations, funding, and corrections retain their recorded status. Slaughter et al. (2026) adds disclosure concerning prior Meta and X/Twitter collaborations; interests warrant scrutiny, not automatic dismissal. "External research" means external to this fellowship, not necessarily independent of providers. Appendix H records the manuscript's targeted checks separately from the source-linked review's counts and depths.

H. Revision scope and source checks · 1 September 2026

Version 1.5 is an editorial and analytical optimisation of v1.4. It clarifies the paired loss-and-gain definition of agency transfer, qualifies the \$10 hook as an unverified budget question, consolidates repeated caveats, and distinguishes actor responsibilities from affected people's rights. The four-part structure, underlying data freezes, quantitative records, 118-implementation review, and production Atlas flags are unchanged. No new experiment, prevalence estimate, independent replication, or double screening is claimed.

Targeted primary-source checks re-examined the Article 50 FAQ and July 2026 amendment, the Commission's ChatGPT designation announcement and register, and the reported outcomes and design limits of the Ibrahim, Poonsiriwong, and Salvi preprints. The

register still lacked the designation decision text. These manuscript checks do not constitute a full current-law or literature audit, reproduce study data, or change the P4 ledger’s 69-source denominator. Binding duties, voluntary guidance, external findings, and proposed controls remain distinct.

References

Source status. Guerrero (2026a–f) identifies materials produced or curated within this programme, not independent corroboration. The empirical results below belong to the cited external studies. “Research literature” does not imply independence from AI providers. Preprints, technical evaluations, internal meta-analyses, observational estimates, corrections, and provider reports retain their status; inclusion is not replication.

Research literature

- Akbulut, Canfer, Rasmi Elasmr, Abhishek Roy, Anthony Payne, Priyanka Suresh, Lujain Ibrahim, Seliem El-Sayed, Charvi Rastogi, Ashyana Kachra, Will Hawkins, Kristian Lum, and Laura Weidinger. 2026. “Evaluating Language Models for Harmful Manipulation.” Preprint, arXiv:2603.25326. <https://arxiv.org/abs/2603.25326>.
- Allcott, Hunt, Juan Camilo Castillo, Matthew Gentzkow, Leon Musolff, and Tobias Salz. 2025. “Sources of Market Power in Web Search: Evidence from a Field Experiment.” NBER Working Paper 33410; revised February 2026. <https://doi.org/10.3386/w33410>.
- Ancina, Amancay, Jana Zumbrägel, and Nicole Krämer. 2026. “I Don’t Believe Misinformation, But I Might Share It: Inoculation Effects in Messenger Communication.” *Social Media + Society* 12 (2). <https://doi.org/10.1177/20563051261449194>.
- Bai, Hui, Jan G. Voelkel, Shane Muldowney, Johannes C. Eichstaedt, and Robb Willer. 2025. “LLM-Generated Messages Can Persuade Humans on Policy Issues.” *Nature Communications* 16: 6037. <https://doi.org/10.1038/s41467-025-61345-5>.
- Biddlestone, Mikey, Jon Roozenbeek, Jane Suiter, Eileen Culloty, and Sander van der Linden. 2025. “Tune in to the Prebunking Network! Development and Validation of Six Inoculation Videos That Prebunk Manipulation Tactics and Logical Fallacies in Misinformation.” *Political Psychology* 46 (6): 1858–1886. <https://doi.org/10.1111/pops.70015>.
- Biega, Asia J., Peter Potash, Hal Daumé III, Fernando Diaz, and Michèle Finck. 2020. “Operationalizing the Legal Principle of Data Minimization for Personalization.” *Proceedings of SIGIR ’20*, 399–408. <https://doi.org/10.1145/3397271.3401034>.
- Chen, Zhongren, Joshua Kalla, Quan Le, Shinpei Nakamura-Sakai, Jasjeet Sekhon, and Ruixiao Wang. 2026. “A Framework to Assess the Persuasion Risks Large Language Model Chatbots Pose to Democratic Societies.” *Journal of Experimental Political Science*, First View. <https://doi.org/10.1017/XPS.2026.10032>.
- Dassanayake, Rishane, Mario Demetroudi, James Walpole, Lindley Lentati, Jason R. Brown, and Edward James Young. 2025. “Manipulation Attacks by Misaligned AI: Risk Analysis and Safety Case Framework.” arXiv:2507.12872. <https://doi.org/10.48550/arXiv.2507.12872>.
- de Freitas Melo, Philipe, Mohamad Hoseini, Savvas Zannettou, and Fabrício Benevenuto. 2024. “Don’t Break the Chain: Measuring Message Forwarding on WhatsApp.” *Proceedings of the International AAAI Conference on Web and Social Media* 18 (1): 1054–1067. <https://doi.org/10.1609/icwsm.v18i1.31372>.
- Debenedetti, Edoardo, Ilia Shumailov, Tianqi Fan, Jamie Hayes, Nicholas Carlini, Daniel Fabian, Christoph Kern, Chongyang Shi, Andreas Terzis, and Florian Tramèr. 2025. “Defeating Prompt Injections by Design.” arXiv:2503.18813v2, 24 June. Preprint. <https://arxiv.org/abs/2503.18813v2>.
- DeVerna, Matthew R., Harry Yaojun Yan, Kai-Cheng Yang, and Filippo Menczer. 2024. “Fact-Checking Information from Large Language Models Can Decrease Headline Discernment.” *Proceedings of the National Academy of Sciences* 121 (50): e2322823121. <https://doi.org/10.1073/pnas.2322823121>.
- Eady, Gregory, Tom Paskhalis, Jan Zilinsky, Richard Bonneau, Jonathan Nagler, and Joshua A. Tucker. 2023. “Exposure to the Russian Internet Research Agency Foreign Influence Campaign on Twitter in the 2016 US Election and Its Relationship to Attitudes and Voting Behavior.” *Nature Communications* 14: 62. <https://doi.org/10.1038/s41467-022-35576-9>.
- El-Sayed, Seliem, Canfer Akbulut, Amanda McCroskery, et al. 2024. “A Mechanism-Based Approach to Mitigating Harms from Persuasive Generative AI.” arXiv:2404.15058. <https://doi.org/10.48550/arXiv.2404.15058>.

- Ershov, Daniel, and Juan S. Morales. 2024. “Sharing News Left and Right: Frictions and Misinformation on Twitter.” *The Economic Journal* 134 (662): 2391–2417. <https://doi.org/10.1093/ej/ueae027>.
- Fang, Cathy Mengying, Auren R. Liu, Valdemar Danry, Eunhae Lee, Samantha W. T. Chan, Pat Pataranutaporn, Pattie Maes, Jason Phang, Michael Lampe, Lama Ahmad, and Sandhini Agarwal. 2025. “How AI and Human Behaviors Shape Psychosocial Effects of Extended Chatbot Use: A Longitudinal Randomized Controlled Study.” arXiv:2503.17473v2, 2 October. Preprint. <https://arxiv.org/abs/2503.17473v2>.
- Gallegos, Isabel O., Chen Shani, Weiyan Shi, Federico Bianchi, Izzy Gainsburg, Dan Jurafsky, and Robb Willer. 2025. “Labeling Messages as AI-Generated Does Not Reduce Their Persuasive Effects.” arXiv:2504.09865v2, 22 April. Preprint. <https://arxiv.org/abs/2504.09865v2>.
- Guess, Andrew M., Brendan Nyhan, and Jason Reifler. 2020. “Exposure to Untrustworthy Websites in the 2016 US Election.” *Nature Human Behaviour* 4: 472–480. <https://doi.org/10.1038/s41562-020-0833-x>.
- Hölbling, Lukas, Sebastian Maier, and Stefan Feuerriegel. 2025. “A Meta-Analysis of the Persuasive Power of Large Language Models.” *Scientific Reports* 15: 43818. <https://doi.org/10.1038/s41598-025-30783-y>.
- Ibrahim, Lujain, Myra Cheng, Cino Lee, Pranav Khadpe, Desmond Ong, Dan Jurafsky, and Diyi Yang. 2026. “Warning Labels Shift Perceptions of Sycophantic AI, but Not Its Influence.” arXiv:2606.21317v1, 19 June. Preprint; version reviewed in the P4 supplement. <https://arxiv.org/abs/2606.21317v1>.
- Kalla, Joshua L., and David E. Broockman. 2018. “The Minimal Persuasive Effects of Campaign Contact in General Elections: Evidence from 49 Field Experiments.” *American Political Science Review* 112 (1): 148–166. <https://doi.org/10.1017/S0003055417000363>.
- Kowal, Matthew, Jasper Timm, Jean-François Godbout, Thomas Costello, Antonio A. Arechar, Gordon Pennycook, David Rand, Adam Gleave, and Kellin Pelrine. 2025. “It’s the Thought that Counts: Evaluating the Attempts of Frontier LLMs to Persuade on Harmful Topics.” arXiv:2506.02873. <https://doi.org/10.48550/arXiv.2506.02873>.
- Lazar, Seth. 2024. “Automatic Authorities: Power and AI.” In *Collaborative Intelligence: How Humans and AI Are Transforming Our World*, edited by Mira Lane and Arathi Sethumadhavan, 37–60. MIT Press. <https://doi.org/10.7551/mitpress/14962.003.0004>.
- Lin, Hause, Gabriela Czarnek, Benjamin Lewis, et al. 2025. “Persuading Voters Using Human–Artificial Intelligence Dialogues.” *Nature* 648: 394–401. <https://doi.org/10.1038/s41586-025-09771-9>.
- Matz, Sandra C., Jacob D. Teeny, Sumer S. Vaid, Heidi Peters, Gabriella M. Harari, and Moran Cerf. 2024. “The Potential of Generative AI for Personalized Persuasion at Scale.” *Scientific Reports* 14: 4692. <https://doi.org/10.1038/s41598-024-53755-0>.
- Moore, Ryan C., Ross Dahlke, and Jeffrey T. Hancock. 2023. “Exposure to Untrustworthy Websites in the 2020 US Election.” *Nature Human Behaviour* 7: 1096–1105. <https://doi.org/10.1038/s41562-023-01564-2>.
- Nouwens, Midas, Ilaria Liccardi, Michael Veale, David Karger, and Lalana Kagal. 2020. “Dark Patterns after the GDPR: Scraping Consent Pop-Ups and Demonstrating Their Influence.” *Proceedings of CHI ’20*, 1–13. <https://doi.org/10.1145/3313831.3376321>.
- Pennycook, Gordon, Ziv Epstein, Mohsen Mosleh, Antonio A. Arechar, Dean Eckles, and David G. Rand. 2021. “Shifting Attention to Accuracy Can Reduce Misinformation Online.” *Nature* 592: 590–595. <https://doi.org/10.1038/s41586-021-03344-2>.
- Pennycook, Gordon, and David G. Rand. 2022. “Accuracy Prompts Are a Replicable and Generalizable Approach for Reducing the Spread of Misinformation.” *Nature Communications* 13: 2333. <https://doi.org/10.1038/s41467-022-30073-5>.
- Pennycook, Gordon, Adam J. Berinsky, Puneet Bhargava, Hause Lin, Rocky Cole, Beth Goldberg, Stephan Lewandowsky, and David G. Rand. 2024. “Inoculation and Accuracy Prompting Increase Accuracy Discernment in Combination but Not Alone.” *Nature Human Behaviour* 8: 2330–2341. <https://doi.org/10.1038/s41562-024-02023-2>.
- Poonsiriwong, Rachel, Chayapatr Archiwanguprok, Constanze Albrecht, Monchai Lertsutthiwong, Pattie Maes, and Pat Pataranutaporn. 2026. “AI Watchdog: Agent Interfaces for Detecting and Defending Against Manipulative Dark Patterns in AI Conversations.” arXiv:2608.21841v1, 22 August. Preprint. <https://arxiv.org/abs/2608.21841v1>.
- Roozenbeek, Jon, Sander van der Linden, Beth Goldberg, Steve Rathje, and Stephan Lewandowsky. 2022. “Psychological Inoculation Improves Resilience against Misinformation on Social Media.” *Science Advances* 8 (34): eabo6254. <https://doi.org/10.1126/sciadv.abo6254>.
- Salvi, Francesco, Manoel Horta Ribeiro, Riccardo Gallotti, and Robert West. 2025. “On the Conversational Persuasiveness of GPT-4.” *Nature Human Behaviour* 9: 1645–1653. <https://doi.org/10.1038/s41562-025-02194-6>.
- Salvi, Francesco, Alejandro Cuevas, and Manoel Horta Ribeiro. 2026. “Commercial Persuasion in AI-Mediated Conversations.” arXiv:2604.04263v1, 5 April. Preprint. <https://arxiv.org/abs/2604.04263v1>.
- Slaughter, Isaac, Axel Peytavin, Johan Ugander, and Martin Saveski. 2025. “Community Notes Reduce Engagement with and Diffusion of False Information Online.” *Proceedings of the National Academy of Sciences* 122 (38): e2503413122. <https://doi.org/10.1073/pnas.2503413122>.

- Slaughter, Isaac, Axel Peytavin, Johan Ugander, and Martin Saveski. 2026. “Correction for Slaughter et al., Community Notes Reduce Engagement with and Diffusion of False Information Online.” *Proceedings of the National Academy of Sciences* 123 (5): e2600635123. Correction to competing-interest disclosure. <https://doi.org/10.1073/pnas.2600635123>.
- Stockwell, Sam. 2024. “AI-Enabled Influence Operations: Threat Analysis of the 2024 UK and European Elections.” CETaS Briefing Papers. Centre for Emerging Technology and Security, The Alan Turing Institute. September 2024. https://cetas.turing.ac.uk/sites/default/files/2024-09/cetas_briefing_paper_-_ai-enabled_influence_operations_-_threat_analysis_of_the_2024_uk_and_european_elections.pdf.
- Susser, Daniel, Beate Roessler, and Helen Nissenbaum. 2019. “Technology, Autonomy, and Manipulation.” *Internet Policy Review* 8 (2): 1–22. <https://doi.org/10.14763/2019.2.1410>.
- Turner, Sarah, July Galindo Quintero, Simon Turner, Jessica Lis, and Leonie Maria Tanczer. 2021. “The Exercisability of the Right to Data Portability in the Emerging Internet of Things (IoT) Environment.” *New Media & Society* 23 (10): 2861–2881. <https://doi.org/10.1177/1461444820934033>.

Law, regulation, standards, and public records

- Associated Press. 2025. “New Hampshire Jury Acquits Consultant Behind AI Robocalls Mimicking Biden on All Charges.” 13 June 2025. <https://apnews.com/article/e87ef8848a2a6515202535a6e800da07>.
- European Commission. 2025a. “Delegated Act on Data Access under the Digital Services Act.” 2 July 2025. <https://digital-strategy.ec.europa.eu/en/library/delegated-act-data-access-under-digital-services-act-dsa>.
- European Commission. 2025b. General-Purpose AI Code of Practice: Safety and Security Chapter. 10 July 2025. <https://ec.europa.eu/newsroom/dae/redirection/document/118119>.
- European Commission. 2026a. “The General-Purpose AI Code of Practice.” Last updated 31 July 2026. <https://digital-strategy.ec.europa.eu/en/policies/contents-code-gpai>.
- European Commission. 2026b. “Third GPAI Signatory Taskforce Meeting – Safety and Security Chapter.” Published 27 April 2026; meeting held 27 March 2026. <https://digital-strategy.ec.europa.eu/en/news/third-gpai-signatory-taskforce-meeting-safety-and-security-chapter>. Accessed 1 September 2026.
- European Commission. 2026c. “Transparency Obligations under Article 50 of the AI Act.” Official explanatory FAQ; read alongside the statutory text and July 2026 amendment. <https://digital-strategy.ec.europa.eu/en/faqs/transparency-obligations-under-article-50-ai-act>. Accessed 1 September 2026.
- European Commission. 2026d. “Commission Designates ChatGPT, Reddit, Roblox under Digital Services Act.” 31 August 2026. <https://digital-strategy.ec.europa.eu/en/news/commission-designates-chatgpt-reddit-roblox-under-digital-services-act>. Accessed 1 September 2026.
- European Commission. 2026e. “Supervision of the Designated Very Large Online Platforms and Search Engines under DSA.” ChatGPT entry, designation dated 31 August 2026; decision text listed as not yet available. <https://digital-strategy.ec.europa.eu/en/policies/list-designated-vlops-and-vloses>. Accessed 1 September 2026.
- European Union. 2022. Regulation (EU) 2022/2065 on a Single Market for Digital Services (Digital Services Act). <https://eur-lex.europa.eu/eli/reg/2022/2065/oj>.
- European Union. 2024a. Regulation (EU) 2024/1689 Laying Down Harmonised Rules on Artificial Intelligence (Artificial Intelligence Act). <https://eur-lex.europa.eu/eli/reg/2024/1689/oj>.
- European Union. 2024b. Regulation (EU) 2024/900 on the Transparency and Targeting of Political Advertising. <https://eur-lex.europa.eu/eli/reg/2024/900/oj>.
- European Union. 2026. Regulation (EU) 2026/1744 of 8 July 2026 Amending Regulations (EU) 2024/1689, (EU) 2018/1139 and (EU) 2023/1230 as Regards Simplification of Implementation. Official Journal, 24 July 2026. In particular Article 1(20), (39) and (40). <https://eur-lex.europa.eu/eli/reg/2026/1744/oj/eng>.
- Federal Communications Commission. 2024a. *Declaratory Ruling: Implications of Artificial Intelligence Technologies on Protecting Consumers from Unwanted Robocalls and Robotexts*, FCC 24-17. <https://docs.fcc.gov/public/attachments/FCC-24-17A1.pdf>.
- Federal Communications Commission. 2024b. *Forfeiture Order: Steve Kramer*, FCC 24-104. <https://docs.fcc.gov/public/attachments/FCC-24-104A1.pdf>.
- Federal Trade Commission. 2019. *Opinion and Final Order against Cambridge Analytica, LLC*. <https://www.ftc.gov/legal-library/browse/cases-proceedings/182-3107-cambridge-analytica-llc-matter>.
- Information Commissioner’s Office. 2018. *Investigation into the Use of Data Analytics in Political Campaigns*. <https://ico.org.uk/media2/migrated/2260271/investigation-into-the-use-of-data-analytics-in-political-campaigns-final-20181105.pdf>.
- National Institute of Standards and Technology. 2024. *Artificial Intelligence Risk Management Framework: Generative Artificial Intelligence Profile*. NIST AI 600-1. <https://doi.org/10.6028/NIST.AI.600-1>.

Provider documentation

- Anthropic. 2026. “Malicious Agentic Influence Campaigns.” In *Claude Mythos Preview System Card*, sec. 8.3.1.3. <https://www-cdn.anthropic.com/53566bf5440a10affd749724787c8913a2ae0841.pdf>.
- Google DeepMind. 2026. *Frontier Safety Framework*, Version 3.1. 17 April 2026. <https://deepmind.google/frontier-safety/>.
- OpenAI. 2026. *Frontier Governance Framework*. 28 May 2026. <https://openai.com/index/openai-frontier-governance-framework/>.

ERA:AI project evidence and research materials

The following programme records support descriptions of the ERA:AI fellowship methods, data and internal coding. They are not independent validation. Records without public links remain on file with the author; access to dual-use material remains subject to ethical and security review. The Policy Evidence Atlas is a public research-preview dataset with separately versioned empirical curation and comparative recommendations.

- Guerrero, Miguel. 2026a. *Agency Transfer Lab: Client-Observed Served-System Evidence Bundles*. Unpublished ERA:AI project evidence; on file with the author.
- Guerrero, Miguel. 2026b. *From Open Weights to Reproducible Intervention: Access-to-Control Methods Record*. Unpublished ERA:AI project evidence; on file with the author.
- Guerrero, Miguel. 2026c. *APE-120 Offline Recovery and Evaluation Registry*. Unpublished ERA:AI project evidence; on file with the author.
- Guerrero, Miguel. 2026d. *Election Evidence Index*. Unpublished ERA:AI project evidence; on file with the author.
- Guerrero, Miguel. 2026e. Agency Transfer Policy Atlas. Public research-preview dataset, v0.1.0-beta.3; comparative recommendation layer v0.4, 1 September 2026. Empirical curation: curation-v0.4-wave1. <https://huggingface.co/datasets/apol/agency-transfer-policy-atlas/tree/v0.1.0-beta.3>.
- Guerrero, Miguel. 2026f. “What Works—and for Whom? P4 Source-Linked Evidence Review, Findings, and Recommendations.” Working research supplement, 1 September 2026. Includes 118 implementation dossiers, 69 source records, a source–implementation crosswalk, and a review-depth protocol. Single-pass author synthesis; not independent replication or a completed causal review of 118 policies. Cambridge ERA project, P4 evidence-review record. Source-linked review (project record).